XIAMEN UNIVERSITY MALAYSIA

ONLINE QUIZ 2 EXAMINATION

Course Code:	EEE313			
Course Name:	Semiconductor Physics and Device Processes			
Question Paper Setter:	MohammadMahdi Ariannejad			
Academic Session:	2021/09	Question Paper:	A ☐	B ☐
Total No. of Pages:	1	Time Allocated:	1 hour	
Additional Materials:	Appendix, Calculator			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. There are three (3) questions. You are assigned to answer all questions.
2. Each question should be answered in detail with proper Solution, only last answer or only formula won't award any marks.
3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1: (30 points)

A one-sided p^+n silicon diode has doping concentrations of $N_a = 5 \times 10^{17}$ cm^{-3} and $N_d = 9 \times 10^{15}$ cm^{-3}. The diode cross-sectional area is $A = 9 \times 10^{-5}$ cm^2.

(***a***) The maximum diffusion capacitance is to be limited to 8 nF. Determine (*i*) the maximum current through the diode, (*ii*) the maximum forward-bias voltage, and (*iii*) the diffusion resistance.

(***b***) Repeat part (*a*) if the maximum diffusion capacitance is limited to 0.98 nF.

(assume $\tau_{p0} = 10^{-7}$ s, $D_p = 10$ cm^2/s)

Question 2: (40 Marks)

An n-channel GaAs pn JFET at $T = 300$ K has parameters $N_d = 6 \times 10^{16}$ cm^{-3}, $N_a = 4 \times 10^{18}$ cm^{-3}, and $a = 0.70\ \mu$m.

(***a***) Calculate the (*i*) internal pinchoff voltage V_{pO} and (*ii*) pinchoff voltage V_p.

(***b***) Determine the minimum undepleted channel thickness, $a - h$, for $V_{GS} = -0.6$ V and for (*i*) $V_{DS} = 0$, (*ii*) $V_{DS} = 0.9$ V, and (*iii*) $V_{DS} = 7.7$ V.

(***c***) Find V_{DS} (sat) for (*i*) $V_{GS} = 0$ and (*ii*) $V_{GS} = -7.0$ V. ($\epsilon_s = (13.1)(8.85\times10^{-14})$, $n_i = 1.8\times10^6$)

Question 3: (30 Marks)

Liu is studying about SOS capacitor as shown in Figure Q2.1. He is assuming that the SiO_2 is ideal (no trapped charge) and has a thickness of $t_{ox} = 800$ Å. The doping concentrations are $N_d = 8\times10^{16}$ cm^{-3} and $N_a = 8\times10^{16}$ cm^{-3}.

(***a***) Calculate the flat-band voltage.

(***b***) Estimate the voltage across the oxide for (*i*) $V_G = +6$ V and (*ii*) $V_G = -6$ V.

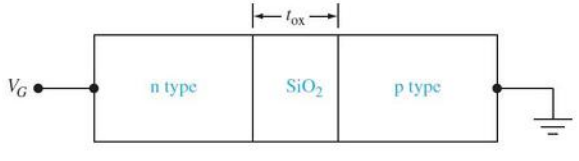


Figure Q2.1.

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